



UIT

UNIVERSITY OF INFORMATION TECHNOLOGY & SCIENCES

Assignment

Course Code-CSE412

Course Title-Operating System Lab

<u>Submitted by-</u>	<u>Submitted to-</u>
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```

#include <iostream>
#include <climits>
using namespace std;

struct Process {
    int at, bt, wt, ct, tat, pos;};

    int quant;
    int main() {
        int n, i, j;

        cout << "Enter no. of processes: ";
        cin >> n;
        Process p[n];

        cout << "Enter quantum: ";
        cin >> quant;

        cout << "Enter process numbers: ";
        for (i = 0; i < n; i++)
            cin >> p[i].pos;

        cout << "Enter arrival time: ";
        for (i = 0; i < n; i++)
            cin >> p[i].at;

        cout << "Enter burst time: ";
        for (i = 0; i < n; i++)
            cin >> p[i].bt;

        int c = n;
        float time = 0, mini = INT_MAX, b[n], a[n];

        int index = -1;
        for (i = 0; i < n; i++) {
            b[i] = p[i].bt;
            a[i] = p[i].at;
        }

        int tot_wt = 0, tot_tat = 0;
        bool flag = false;

        while (c != 0) {
            mini = INT_MAX;
            flag = false;

            for (i = 0; i < n; i++) {
                float p = time + 0.1;
                if (a[i] <= p && mini > a[i] && b[i] > 0) {
                    index = i;
                    mini = a[i];
                    flag = true;
                }

                if (!flag) {
                    time++;
                    continue;
                }

                if (b[index] <= quant) {
                    time += b[index];
                    b[index] = 0;
                } else {
                    time += quant;
                    b[index] -= quant;
                }

                if (b[index] > 0) {
                    a[index] = time + 0.1;
                }

                if (b[index] == 0) {c--;
                    p[index].ct = time;
                    p[index].wt = p[index].ct - p[index].at - p[index].bt;
                    tot_wt += p[index].wt;
                    p[index].tat = p[index].bt + p[index].wt;
                    tot_tat += p[index].tat;
                }
            }

            cout << "P\tAT\tBT\tCT\tTAT\tWT" << endl;

            for (i = 0; i < n; i++) {
                cout << p[i].pos << "\t";
                cout << p[i].at << "\t";
                cout << p[i].bt << "\t";
                cout << p[i].ct << "\t";
                cout << p[i].tat << "\t";
                cout << p[i].wt << endl;
            }

            // Calculating average wait time and turnaround time
            double avg_wt = tot_wt / static_cast<double>(n);
            double avg_tat = tot_tat / static_cast<double>(n);

            cout << "Avg WT: " << avg_wt << endl;
            cout << "Avg TAT: " << avg_tat << endl;

            return 0;
        }
    }

```

main.c...

Output

```

Enter no. of processes: 5
Enter quantum: 3
Enter process numbers: 1 2 3 4 5
Enter arrival time: 2 1 4 0 6
Enter burst time: 6 3 2 5 4

```

P	AT	BT	CT	TAT	WT
1	2	6	19	17	11
2	1	3	6	5	2
3	4	2	13	9	7
4	0	5	11	11	6
5	6	4	20	14	10

```

Avg WT: 7.2
Avg TAT: 11.2

=== Code Execution Successful ===

```